

WEEK 3 REPORT:MACHINE LEARNING FUNDAMENTALS

Introduction

During Week 3 of the AnalystLab Africa Machine Learning Internship Program, I was introduced to the fundamental concepts involved in machine learning workflows. Using the Titanic dataset, I explored how machine learning problems are structured, how data is prepared for modelling, and how model performance can be evaluated.

The main concepts covered during this week included supervised and unsupervised learning, train-test splitting, overfitting and under fitting, and basic model evaluation techniques.

Lessons Learned

This week helped me understand the difference between supervised and unsupervised learning. I learned that supervised learning uses labelled data, where the model learns from known outcomes to make predictions. In the Titanic dataset, the survival outcome acts as the target variable, while passenger characteristics are used as predictor variables.

I also learned the importance of dividing datasets into training and testing sets. The training set allows a machine learning model to learn patterns from existing data, while the testing set evaluates whether the model can make accurate predictions on unseen data. This process is important because it helps measure how well a model can generalize beyond the data it was trained on.

Another key concept I learned was the difference between overfitting and under fitting. Overfitting occurs when a model learns the training data too closely and performs poorly on new data, while under fitting occurs when a model is too simple to capture important patterns. Understanding these concepts showed me the importance of finding a balance when developing machine learning models.

Finally, I learned about basic model evaluation techniques, including accuracy, confusion matrices, precision, and recall. These evaluation methods help determine whether a classification model is performing effectively and provide insight into the types of errors made by the model.

Challenges

One of the challenges I encountered was understanding how machine learning concepts connect with practical implementation. Although the mathematical and statistical ideas behind machine learning are familiar, translating these concepts into a machine learning workflow required deeper understanding.

Another challenge was understanding the importance of evaluating models using appropriate metrics rather than relying only on accuracy. Different evaluation measures provide different perspectives on model performance, especially when dealing with classification problems.

Key Insights

This week highlighted that machine learning involves more than simply applying algorithms to data. A successful machine learning workflow requires understanding the data, selecting appropriate features, dividing data correctly, and evaluating whether the model can generalize to new observations.

I gained a better understanding of how statistical concepts are applied in machine learning and how careful evaluation helps build more reliable predictive models. The concepts learned this week provide a foundation for exploring more advanced machine learning techniques in future projects.

Conclusion

The Week 3 Machine Learning Fundamentals module improved my understanding of the overall machine learning process. The knowledge gained from supervised learning, data splitting, model performance evaluation, and model limitations will be valuable in future data analysis and machine learning projects.